Comprehensive Exam-Oriented Study Guide

DATA AND VISUAL ANALYTICS (AL-603-B)

UNIT III

Data Wrangling

B.Tech, VI Semester | RGPV Examination Scheme (GS)

DATA AND VISUAL ANALYTICS (AL-603-B)

Syllabus Topics Covered in This Unit
Introduction to Data Wrangling
Gathering Data
Assessing Data
Cleaning Data

Table of Contents

Unit Overview & Syllabus Map

Unit III is intentionally compact but conceptually critical: real-world data is almost never analysis-ready the moment it is collected. Data Wrangling (also called data munging) is the practical, hands-on discipline of taking raw, messy, inconsistent data and transforming it into a clean, structured, reliable form that statistical techniques (Units I-II) and visualization tools (Units IV-V) can actually use. Analysts routinely report spending 60-80% of total project time on wrangling — making this one of the highest real-world-value units in the entire syllabus, even though it has fewer sub-topics than Units I and II.

Official Syllabus Text

DEFINITION

Data Wrangling: Intro to Data Wrangling, Gathering Data, Assessing Data, Cleaning Data.

LEARNING OBJECTIVES — By the end of this unit, you should be able to:

✓ Define data wrangling and describe the full Gather → Assess → Clean workflow, plus the extended six-stage industry lifecycle.

✓ List and evaluate different sources of data, including the primary vs secondary data distinction.

✓ Distinguish 'dirty' data issues from 'messy' (structural/tidiness) data issues when assessing a dataset.

✓ Apply appropriate techniques to handle missing values, duplicates, outliers, and inconsistent formatting during cleaning.

1. Introduction to Data Wrangling

1.1 Introduction

Data Wrangling is the process that sits between 'raw data exists somewhere' and 'my model/dashboard/report can use this data reliably.' It is not a single technique but an end-to-end workflow combining data collection, structural transformation, quality assessment, and cleaning — all aimed at converting messy, real-world data into a dependable, analysis-ready dataset.

1.2 Definition

DEFINITION

Data Wrangling (Data Munging): The process of gathering, structuring, cleaning, and enriching raw data from one or more sources into a well-organized, consistent, and analysis-ready format, so that it can be used reliably for statistical analysis, visualization, or machine learning.

1.3 Why Data Wrangling Is Needed — The Core Problem

Data collected from the real world is almost always imperfect. Common problems include:

- Data spread across multiple disconnected sources (databases, spreadsheets, APIs, logs, PDFs).
- Inconsistent formats (dates as 'DD/MM/YYYY' in one file and 'Month DD, YYYY' in another).
- Missing values, duplicate records, and typographical errors.
- Irrelevant columns/rows that add noise rather than insight.
- Data not structured in the 'tidy' row-per-observation, column-per-variable format that most tools expect.

Data Wrangling systematically addresses every one of these problems before any meaningful analysis begins.

1.4 The Data Wrangling Workflow — Overview

While the RGPV syllabus names three core stages (Gathering, Assessing, Cleaning), the broader industry-standard data wrangling lifecycle (as popularized by tools like Trifacta and standard DVA textbooks) extends this into six stages. Knowing this fuller picture lets you write a more complete, higher-scoring answer.

```
DISCOVERING -> STRUCTURING -> CLEANING -> ENRICHING -> VALIDATING -> PUBLISHING
(Gather &      (Reshape into    (Fix errors,    (Add/merge     (Quality &    (Deliver to
 understand      tidy rows/       missing,        useful         consistency    analysts /
 the data)       columns)         duplicates)     external       checks)        models /
                                                   data)                        dashboards)
```

1. Discovering / Gathering: Locate, collect, and get a first, high-level understanding of the available data.
2. Structuring: Reorganize the raw data into a consistent, usable structure (e.g., converting nested JSON into flat tables, reshaping wide data into long/tidy format).
3. Cleaning: Fix or remove errors, missing values, duplicates, and inconsistent formatting.
4. Enriching: Augment the dataset by adding new derived columns, or joining it with additional external/internal data sources for more context.

5. Validating: Run consistency, quality, and integrity checks to confirm the data is now trustworthy (e.g., verifying data types, ranges, uniqueness constraints).
6. Publishing: Deliver the finalized, wrangled dataset to its destination — a database, an analytics tool, a dashboard, or a machine learning pipeline.

EXAM TIP

RGPV's syllabus names only 3 stages (Gathering, Assessing, Cleaning) — always cover those three in full depth first, then mention Structuring, Enriching, Validating, and Publishing as 'additional/extended stages in the full industry wrangling lifecycle' for bonus depth and marks.

1.5 Key Concepts

- Tidy Data:
- A structuring principle where every variable is a column, every observation is a row, and every type of observational unit forms its own table — the ideal target structure for wrangling.
- Data Quality:
- The degree to which data is accurate, complete, consistent, timely, and valid for its intended use — the outcome wrangling aims to maximize.
- ETL (Extract, Transform, Load):
- A closely related data-engineering process; data wrangling is often considered the more exploratory, analyst-driven cousin of formal ETL pipelines.
- Data Provenance/Lineage:
- Tracking where data came from and what transformations were applied to it — important for reproducibility and trust.

1.6 Tools Commonly Used for Data Wrangling

Tool / Library	Category	Typical Use
Python Pandas	Programming library	Reading, reshaping, cleaning, and merging tabular data
OpenRefine	Standalone GUI tool	Interactive cleaning of messy, inconsistent categorical data
Trifacta / Dataprep	Commercial wrangling platform	Visual, guided wrangling with automated suggestions
Microsoft Excel / Power Query	Spreadsheet tool	Manual and semi-automated cleaning for smaller datasets
SQL	Query language	Filtering, joining, and aggregating data directly in a database
R (dplyr, tidyr)	Programming library	Tidying and transforming data within the R ecosystem

1.7 Real-World Example

An e-commerce company wants to analyze customer purchase behaviour. Raw data arrives as: a MySQL 'orders' table, a CSV export of customer support tickets, and a JSON feed from a third-party marketing API. Data wrangling gathers all three sources, restructures the JSON into flat tables, standardizes customer IDs across systems, removes duplicate/test orders, fills in missing shipping regions, and finally publishes one unified 'customer_360' table ready for the analytics team.

1.8 Advantages of Proper Data Wrangling

- Improves the accuracy and reliability of all downstream analysis and models — 'garbage in, garbage out' is prevented at the source.
- Saves significant time later in the project, since clean, well-structured data is far faster to analyze and visualize.
- Enables data from multiple disparate sources to be combined meaningfully.
- Builds trust in reported insights and dashboards among business stakeholders.

1.9 Disadvantages / Challenges

- Extremely time-consuming — commonly cited as 60-80% of a data project's total effort.
- Requires strong domain knowledge to correctly judge what counts as an 'error' versus a legitimate unusual value.
- Manual wrangling steps can be difficult to reproduce and audit unless carefully scripted and documented.
- Poorly wrangled data can introduce NEW biases or errors if cleaning rules are applied carelessly.

1.10 Applications

- Preparing raw survey data for statistical analysis.
- Consolidating multi-source business data into a single warehouse/dashboard.
- Preprocessing data before training any machine learning model.
- Preparing government/public datasets for open-data publishing.

MEMORY TIP

Data Wrangling memory tip: 'GAC — Gather, Assess, Clean' for the RGPV syllabus core; extend to 'DSCEVP' (Discover, Structure, Clean, Enrich, Validate, Publish) for the full industry lifecycle.

2. Gathering Data

2.1 Introduction

Gathering Data is the first practical stage of data wrangling — locating, collecting, and importing raw data from all relevant sources into a working environment where it can be examined and processed. The quality and completeness of this stage directly determines what analysis is even possible later, making it a foundational step rather than a mere formality.

2.2 Definition

DEFINITION

Data Gathering: The process of identifying, collecting, and importing raw data from one or more internal or external sources into an analysis environment, in preparation for subsequent assessment and cleaning.

2.3 Common Sources of Data

Source Category	Examples
Databases	Relational (MySQL, PostgreSQL, Oracle), NoSQL (MongoDB, Cassandra)
Files	CSV, Excel, JSON, XML, plain text, log files
Web-based sources	APIs (REST/SOAP), web scraping, RSS feeds
Enterprise systems	CRM, ERP, HR systems, transactional/POS systems
Public/Open data	Government portals, Kaggle datasets, census data, research repositories
Sensors/IoT	Machine logs, GPS trackers, industrial sensor streams
Surveys and manual entry	Google Forms, paper survey digitization, interview transcripts

2.4 Primary vs Secondary Data (Key Distinction)

Aspect	Primary Data	Secondary Data
Definition	Data collected firsthand by the analyst/organization for a specific purpose	Data collected earlier by someone else, reused for a new purpose
Examples	A survey you personally design and run	A government census report, a purchased dataset
Advantage	Highly relevant, controlled quality, up to date	Cheaper and faster to obtain; often larger in scale
Disadvantage	Time-consuming and expensive to collect	May not perfectly fit the current need; quality/recency uncertain

2.5 Common Data Gathering Techniques / Methods

- Database querying:
- Using SQL to extract relevant tables/records directly from a relational database.
- API calls:
- Programmatically requesting data from a web service in JSON/XML format (e.g., a weather API, a social media API).
- Web scraping:
- Automatically extracting data from web pages when no API is available (using tools like BeautifulSoup, Scrapy).
- File import:
- Reading existing CSV, Excel, or JSON files into an analysis tool (e.g., pandas.read_csv()).
- Surveys and forms:
- Directly collecting primary data from respondents.
- Manual data entry / digitization:
- Converting physical documents or unstructured notes into digital records.

2.6 Working Process — Step-by-Step Data Gathering Methodology

1. Clarify the analytical question/goal — this determines exactly what data is actually needed.
2. Identify all potential internal and external data sources that could answer the question.
3. Evaluate each source for relevance, accessibility, cost, and licensing/legal permissions.
4. Extract/download/query the data using the appropriate method (SQL, API call, file download, scraping, survey).
5. Perform an initial 'landing' of the raw data into a working environment (e.g., a raw/staging folder or database schema), preserving the original untouched copy for traceability.
6. Document metadata: source, collection date, format, and any known limitations of each dataset gathered.

2.7 Real-World Example

A retail analytics team gathering data for a customer churn study pulls: (a) transaction history from the company's SQL sales database, (b) customer support ticket data via a CRM's REST API, and (c) regional demographic data from a government open-data portal (secondary data) — combining primary and secondary sources into one gathering stage.

2.8 Advantages of a Careful Gathering Process

- Ensures analysis is based on complete and relevant data rather than whatever happens to be conveniently available.
- Establishes traceability/provenance from the very start of the project.
- Combining multiple sources early enables richer, more powerful analysis later.

2.9 Disadvantages / Challenges

- Data access can be restricted by permissions, cost, rate limits (APIs), or legal/privacy regulations.
- Sources may use incompatible formats and identifiers, complicating later integration.
- Web scraping can be fragile (breaks when website structure changes) and may violate terms of service if done carelessly.

2.10 Applications

- Building a unified customer data warehouse from multiple business systems.
- Collecting training data for machine learning models from public and proprietary sources.
- Aggregating social media data for sentiment analysis via APIs.

MEMORY TIP

Gathering memory tip: 'DBAFS' — Databases, Bulk files, APIs, Forms/Surveys, Scraping — the five classic sources to name in any Gathering Data answer.

EXAM TIP

When asked about Data Wrangling broadly, spend roughly 2 of your 7 marks explicitly on Gathering Data — name at least 4 different source types and mention the Primary vs Secondary distinction; RGPV examiners specifically look for this breadth.

3. Assessing Data

3.1 Introduction

Once data has been gathered, it cannot be assumed to be correct or usable — it must be systematically examined to discover exactly what quality issues exist before any cleaning is attempted. Assessing Data is this diagnostic stage: profiling the dataset to identify missing values, inconsistencies, duplicates, outliers, and structural problems.

3.2 Definition

DEFINITION

Data Assessment (Data Profiling): The process of systematically examining a gathered dataset to identify its structure, content, and quality issues — including missing values, duplicates, inconsistent formats, outliers, and structural problems — in order to plan the appropriate cleaning steps.

3.3 Two Types of Data Quality Issues (Key Conceptual Distinction)

Issue Type	Definition	Example
Dirty Data (Quality Issues)	Content-level problems — the data is present but factually or formatly wrong	Missing values, duplicate rows, invalid entries, outliers, inconsistent capitalization
Messy Data (Tidiness/Structural Issues)	Structural problems — the data is arranged in a way that makes it hard to analyze, even if the values themselves are correct	Column headers that are actually values, multiple variables stored in one column, one variable spread across multiple columns

This distinction (popularized by data scientist Hadley Wickham's 'Tidy Data' framework) is an excellent point to raise in exam answers, since it shows depth beyond just 'find missing values.'

3.4 Common Data Quality Dimensions Assessed

- Completeness:
- Are there missing values? What proportion of each column is empty/null?
- Validity:
- Do values conform to the expected format, type, or range (e.g., age should not be negative, email should match a valid pattern)?
- Accuracy:
- Do the values correctly reflect real-world facts (e.g., is the recorded city name actually correct for that pin code)?
- Consistency:
- Are values represented uniformly across the dataset (e.g., 'USA', 'U.S.A.', and 'United States' all meaning the same thing)?
- Uniqueness:
- Are there duplicate records that should be a single entry?
- Timeliness:
- Is the data current enough to be useful for the intended analysis?

3.5 Methods and Techniques for Assessing Data

- Visual assessment:
- Manually scrolling through and eyeballing a sample of the dataset (spreadsheet view, head()/tail() commands) to spot obvious problems.
- Programmatic assessment:
- Using code (e.g., pandas .info(), .describe(), .isnull().sum(), .duplicated()) to systematically summarize data types, ranges, missing counts, and duplicate counts.
- Summary statistics:
- Using descriptive statistics (Unit I) — mean, median, min, max, standard deviation — to spot impossible or extreme values.
- Visualization-based assessment:
- Histograms, box plots, and scatter plots to visually spot outliers, skew, and unusual clusters.
- Cross-field / business-rule validation:
- Checking logical relationships between columns (e.g., 'ship date' should never be before 'order date').

3.6 Working Process — Step-by-Step Data Assessment Methodology

1. Get an overview: check dataset shape (rows × columns), data types of each column, and a sample of records.
2. Assess completeness: quantify missing/null values per column and per row.
3. Assess validity: check that each column's values match its expected type, format, and permissible range.
4. Assess consistency: look for the same real-world entity represented in multiple different ways (e.g., inconsistent category spellings).
5. Assess uniqueness: check for exact and near-duplicate records.
6. Assess accuracy: cross-check a sample of records against a trusted external source, where feasible.
7. Document every issue found in a structured 'data quality issue list', tagging each as 'dirty' (content) or 'messy' (structural), ready to feed into the Cleaning stage.

3.7 Diagram — Assessment Flow

```
Gathered Raw Data
       |
       v
+                             +
|  Visual scan  + programmatic |
|  profiling (types, nulls,    |
|  summary stats, duplicates)  |
+                             +
       |
       v
Documented list of issues:
- Dirty data issues  (missing, invalid, duplicate, outlier)
- Messy data issues  (structural / tidiness problems)
       |
       v
   Feeds into -> CLEANING stage
```

3.8 Real-World Example

Assessing a gathered customer dataset reveals: 12% of 'phone_number' values are missing (completeness issue), some 'age' values are negative (validity issue), 'country' contains both 'India' and 'IN' for the same value (consistency issue), 340 exact duplicate customer rows exist (uniqueness issue), and the 'address' column actually contains street + city + pincode combined together (a messy/structural tidiness issue).

3.9 Advantages of Rigorous Assessment

- Prevents wasted effort cleaning the wrong things by first precisely identifying real issues.
- Creates a clear, prioritized roadmap for the cleaning stage.
- Surfaces structural (messy) problems early, before they silently break downstream calculations.

3.10 Disadvantages / Challenges

- Manual visual assessment does not scale to very large datasets (millions of rows).
- Some issues (like accuracy) require external validation sources that may not be available.
- Judging what counts as an 'outlier' vs a genuine unusual-but-valid value requires domain expertise, introducing subjectivity.

3.11 Applications

- Pre-analysis data quality audits in any BI/analytics project.
- Data governance and compliance reporting (data quality dashboards).
- Preprocessing checks before feeding data into machine learning pipelines.

MEMORY TIP

Assessment memory tip: 'CVACUT' — Completeness, Validity, Accuracy, Consistency, Uniqueness, Timeliness — the six dimensions to check every time you assess a dataset.

EXAM TIP

Always explicitly mention the Dirty vs Messy data distinction when answering an Assessing Data question — it is a well-known DVA textbook framework that most students miss, and naming it signals stronger conceptual depth to the examiner.

4. Cleaning Data

4.1 Introduction

Cleaning Data is the corrective stage that directly acts on the issues identified during assessment — fixing, removing, or otherwise handling missing values, duplicates, invalid entries, inconsistent formats, and structural problems so the dataset finally becomes trustworthy and analysis-ready. This is usually the most hands-on and time-intensive stage of the entire wrangling process.

4.2 Definition

DEFINITION
Data Cleaning (Data Cleansing): The process of detecting and correcting (or removing) inaccurate, incomplete, duplicate, inconsistent, or structurally malformed records within a dataset, in order to improve its overall quality and fitness for analysis.

4.3 Major Categories of Cleaning Operations

(a) Handling Missing Values

Technique	Description	When to Use
Deletion (listwise/row)	Remove entire rows containing missing values	When missing data is a small % and random
Deletion (column)	Remove an entire column if too much of it is missing	When a column is mostly empty and not critical
Mean/Median/Mode Imputation	Replace missing values with the column's mean, median, or mode	Quick fix for numeric (mean/median) or categorical (mode) data with few missing values
Forward/Backward Fill	Carry the previous/next valid value forward/backward	Time-series data where values change gradually
Predictive Imputation	Use a model (e.g., regression, k-NN) to predict the missing value from other columns	When missingness is significant and other columns are informative
Flagging	Keep missing values but add an indicator column marking them as missing	When the fact that data is missing is itself meaningful

(b) Removing Duplicates

Identify and remove exact or near-duplicate records (e.g., the same customer entered twice with slightly different spelling), typically using unique-key matching or fuzzy-matching techniques for near-duplicates.

(c) Correcting Structural Errors

- Fixing inconsistent capitalization/spacing (e.g., ' delhi', 'Delhi ', 'DELHI' → 'Delhi').
- Standardizing categorical labels (e.g., 'M', 'Male', 'male' → a single consistent 'Male').
- Correcting mislabeled data types (e.g., a numeric column stored as text).
- Resolving structural/tidiness issues identified in assessment (e.g., splitting a combined 'address' column into street, city, pincode).

(d) Handling Outliers

- Verify whether an extreme value is a genuine data entry error or a legitimate rare event.
- Common statistical rules: values beyond 1.5×IQR from Q1/Q3 (box-plot rule), or beyond 3 standard deviations from the mean (z-score rule).
- Decision options: correct the value if it's a clear error, remove the record, or keep it but flag it (especially if it's a genuine business-relevant extreme case, like a fraud outlier).

(e) Handling Irrelevant or Invalid Data

- Removing columns/rows that are not relevant to the analysis objective.
- Filtering out invalid entries that violate business rules (e.g., a 'delivery date' before the 'order date').

(f) Standardization and Normalization (Format-Level Cleaning)

- Standardizing date formats to a single consistent format (e.g., ISO 8601: YYYY-MM-DD).
- Standardizing units of measurement (e.g., converting all weights to kilograms).
- Trimming whitespace and normalizing text encoding.

4.4 Working Process — Step-by-Step Data Cleaning Methodology

1. Take the prioritized issue list produced during the Assessing stage as the cleaning plan.
2. Define and apply a cleaning rule for each issue (e.g., 'impute missing age with median age', 'drop exact duplicate rows', 'standardize country names using a lookup table').
3. Apply structural/'messy data' fixes first (reshaping, splitting/merging columns) since these often affect how later dirty-data fixes should be applied.
4. Apply dirty-data fixes: handle missing values, remove duplicates, correct invalid/inconsistent entries, and treat outliers.
5. Re-run the assessment checks (Section 3) after cleaning to verify the issues are actually resolved — cleaning is iterative, not a single pass.
6. Document every cleaning transformation applied, preserving the ability to trace back to the original raw data (data lineage).
7. Finalize and 'publish' the cleaned dataset for the next stage of analysis or modelling.

```
Raw Data  --(Gather)-->  Collected Data  --(Assess)-->  Issue List
                                                             |
                                                             v
                                                   +-----------------  +
                                                   |     CLEAN         |
                                                   | - Fix structure   |
                                                   | - Fix missing     |
                                                   | - Remove dupes    |
                                                   | - Fix invalid     |
                                                   | - Handle outliers|
                                                   +-------------------+
                                                             |
                                                             v
                                              Clean, Analysis-Ready Dataset
```

4.5 Real-World Example

Continuing the customer dataset example from Section 3: missing phone numbers are imputed where possible or flagged; negative age values are corrected where a data-entry sign error is evident, otherwise the row is removed; 'India'/'IN' entries are standardized to a single consistent label using a lookup table; the 340 exact duplicate rows are dropped; and the combined 'address' column is split into separate street, city, and pincode columns — producing a clean, structured 'customer_master' table ready for churn analysis.

4.6 Advantages of Thorough Data Cleaning

- Directly improves the accuracy and trustworthiness of every downstream statistical test, model, and visualization.
- Reduces the risk of misleading conclusions caused by errors, duplicates, or inconsistent categories.
- Enables reliable joining/merging of data from multiple sources once formats are standardized.

4.7 Disadvantages / Limitations

- Imputation and outlier removal can introduce their own bias if not done carefully (e.g., mean imputation understates true variability).
- Extremely time-consuming, especially for large, highly inconsistent datasets.
- Over-aggressive cleaning risks discarding genuinely valid, informative data (e.g., real rare events mistaken for outliers).
- Requires ongoing maintenance, since new data arriving later can reintroduce the same quality issues.

4.8 Applications

- Preparing data for machine learning model training (most ML algorithms perform poorly on dirty data).
- Ensuring accurate KPI reporting in business intelligence dashboards.
- Regulatory compliance and audit-readiness of enterprise data.
- Master Data Management initiatives that maintain a single, clean version of core business entities.

MEMORY TIP

Cleaning memory tip: 'MDICO' — Missing values, Duplicates, Invalid entries, Consistency (formatting/labels), Outliers — the five things to explicitly clean in any answer.

EXAM TIP

For 'Explain Data Wrangling process in detail' (the exact May 2024 wording), give roughly equal weight to Gathering, Assessing, and Cleaning (about 2-2-3 marks worth of content), and always close with the end-to-end flow diagram (Gather → Assess → Clean → Analysis-ready data) — this single diagram visually demonstrates full syllabus coverage to the examiner.

5. Exam Focus & Quick Revision

5.1 Most Important Topics (Ranked by PYQ Frequency)

1. Data Wrangling — overall definition and full process (Gather → Assess → Clean) — asked directly in BOTH 2023 and 2024, the single most important topic in this unit.
2. Cleaning Data techniques (missing values, duplicates, outliers, standardization) — the most detail-heavy sub-topic and most likely to carry follow-up marks.
3. Assessing Data (data quality dimensions, dirty vs messy data) — important conceptual differentiator that boosts answer depth.
4. Gathering Data (sources, primary vs secondary data) — smaller topic but easy, guaranteed marks if a sub-question specifically asks about sources.

5.2 Expected Long-Answer Questions (7 / 14 Marks)

- What do you mean by Data Wrangling? Explain the Data Wrangling process in detail. (asked verbatim, May 2024)
- Discuss in detail about Data Wrangling. (asked verbatim, May 2023)
- Explain the various stages of data wrangling with a suitable diagram.
- What are the different sources of data used in the data gathering stage? Differentiate primary and secondary data.
- Explain data assessment. What is the difference between dirty data and messy data?
- Explain different techniques for cleaning data, especially for handling missing values and outliers.
- Explain the complete data wrangling lifecycle: Discovering, Structuring, Cleaning, Enriching, Validating, Publishing.

5.3 Expected Short-Answer / Definition Questions

- Define data wrangling.
- List any four sources of data.
- Differentiate primary and secondary data.
- What is dirty data? What is messy/tidy data?
- Name any three techniques for handling missing values.
- What is an outlier? Name one method to detect it.
- Define data cleaning.

5.4 Frequently Repeated Concepts to Master First

- The Gather → Assess → Clean process flow, with the extended Discover/Structure/Clean/Enrich/Validate/Publish lifecycle as a bonus.
- Dirty data vs messy (tidiness) data distinction.
- Six data quality dimensions: Completeness, Validity, Accuracy, Consistency, Uniqueness, Timeliness.
- Missing value handling techniques table (deletion, imputation, fill, predictive, flagging).
- Outlier detection rules (IQR rule, z-score rule).

5.5 Quick Revision Sheet

Concept	One-Line Definition / Key Point
Data Wrangling	Gathering, structuring, cleaning, and enriching raw data into an analysis-ready format
Gathering Data	Locating and collecting raw data from internal/external sources
Primary Data	Collected firsthand for a specific purpose
Secondary Data	Collected earlier by someone else, reused for a new purpose
Assessing Data	Profiling data to identify quality and structural issues before cleaning
Dirty Data	Content-level issues: missing, invalid, duplicate, outlier values
Messy Data	Structural/tidiness issues: wrong shape, combined columns, etc.
Tidy Data	Every variable a column, every observation a row, every unit its own table
Cleaning Data	Fixing/removing errors, duplicates, missing values, and structural problems
Imputation	Filling in missing values using mean/median/mode/model-based prediction
Outlier	A value far outside the normal range — detected via IQR or z-score rules
Full Wrangling Lifecycle	Discover → Structure → Clean → Enrich → Validate → Publish

5.6 Common Questions Linked to This Unit

- Walk me through how you would clean a messy real-world dataset from scratch.
- How do you decide whether to delete or impute missing values?
- How do you detect and handle outliers, and how do you decide if they're errors or real events?
- What is the difference between data cleaning and data wrangling?
- How would you handle data coming from three different systems with different customer ID formats?
- What tools have you used for data wrangling, and why did you choose them?

End of Unit III Study Guide. Cross-check each topic above against the latest official RGPV syllabus copy before your exam, as syllabi are occasionally revised.